



HUANGKUIWUFANG inhibits pyruvate dehydrogenase to improve glomerular injury in anti-Thy1 nephritis model

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ABSTRACT

Ethnopharmacological relevance: Huangkuiwufang (HKSFW) is composed of *Abelmoschus manihot* (L.) Medik., *Astragalus mongolicus*, *Polygonum cuspidatum*, *Curcuma longa* L. *Abelmoschus Manihot* (L.) Medik. has been widely used for the treatment of chronic renal disease, oral ulcers and burn in China for centuries (Committee of the Pharmacopoeia of PR China, 2010). *Abelmoschus manihot* (L.) Medik., *Polygonum cuspidatum*, *Curcuma longa* L. have been mainly applied in folk medicine for their therapeutic effects on diabetes, cancer, heart disease and other diseases.

Aim of the study: We aimed to investigate the renoprotective function of HKSFW in anti-Thy nephritis model and clarify the relevant mechanisms.

Materials and methods: One week after the model of glomerulonephritis created by injecting anti-thymocyte serum (ATS), rats were treated with Huangkui capsule, enalapril or HKSFW by gavage for a period of 8 weeks. The therapeutic effect was evaluated by detection of proteinuria, plasma creatine, blood urea nitrogen (BUN), podocyte injury, glomerular accumulation of extracellular matrix (ECM) and the markers of oxidative stress and renal fibrosis. RNA Sequencing (RNA-seq), KEGG and western blotting analysis were performed to indicate the signaling pathway involved in the therapeutic effect of HKSFW.

Results: Nephritic rats presented the increase of BUN, serum creatinine (Scr), proteinuria, podocyte damage, glomerular fibrosis, Ang II type 1 receptor (AT1R), and the reduction of creatinine clearance (Ccr). In contrast, application of HKSFW to nephritic rats decreased the levels of BUN and proteinuria, promoted mesangial cell recovery and improved oxidative stress level and podocyte injury. KEGG analysis revealed that pyruvate metabolism was the most significantly upregulated pathway in rats treated with HKSFW compared to disease control group. Increased pyruvate dehydrogenase and PAI-1 caused by nephritis was inhibited by HKSFW interposition. Furthermore, dichloroacetate sodium (DCA), an agonist of pyruvate dehydrogenase, could stimulate PAI-1 expression, which was suppressed by HKSFW.

Conclusion: Chinese herbal preparation HKSFW has remarkable curative effects on glomerulonephritis animals. HKSFW attenuates pyruvate dehydrogenase to improve glomerular injury.

1. Introduction

Kidney disease is a common disease that seriously endangers the

health of mankind, mainly including different types of nephritis, like acute renal failure, kidney stones, renal cysts and so on (Eckardt et al., 2013). The worldwide prevalence of chronic kidney disease (CKD) is

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Abbreviations

ATS	Anti-thymocyte serum
BUN	Blood urea nitrogen
BP	Biological process
CC	Cellular component
Ccr	Creatinine clearance
CG	Chronic glomerulonephritis
CKD	Chronic kidney disease
Cr	creatinine
DAG	Directed acyclic graph
DC	disease control
DCA	Dichloroacetate sodium
DEGs	Differentially expressed genes
DMEM	Dulbecco's modified Eagle's medium
ECM	Extracellular matrix
Ena	Enalapril
FBS	Fetal bovine serum

GAPDH	Glyceraldehyde-3-phosphate dehydrogenase
HKSWF	Huangkuisiwufang
MF	Molecular function
MMC	murine mesangial cells
NC	Normal control group
Nrf2	nuclear-factor-erythroid 2-related factor
PAGE	Polyacrylamide gel electrophoresis
PAI-1	Plasminogen activator inhibitor-1
PAS	Periodic acid Schiff
PBS	Phosphate-buffered saline
PDC	Pyruvate dehydrogenase complex
PVDF	Poly (vinylidene fluoride)
RNA-seq	RNA Sequencing
Scr	serum creatinine
SD	Sprague-Dawley
SDS	Sodium dodecyl sulfate
TCM	Traditional Chinese medicine

estimated at 8–16% (Jha et al., 2013). Clinically, CKD is characterized by decreased glomerular filtration rate and increased urinary albumin excretion (Vroiling et al., 2016). Chronic glomerulonephritis (CG), as one of the main causes of CKD, presents various etiologies and different pathological morphologies (Chiu et al., 2015). CG, referred to as chronic nephritis, is a glomerulonephritis with proteinuria, hematuria, hypertension and edema as the basic clinical manifestations (Natarajan and Jeyachandran, 2014). It develops in different ways with slow progression and varying degrees of renal dysfunction, which eventually leads to chronic renal failure (Liu et al., 2016). The progression of CG is accompanied by the occurrence of renal fibrosis, and can be aggravated by the injury of podocyte (Morioka and Ikezumi, 2001). Additionally, mesangiolysis could occur in humans as well as in experimental animal models of kidney disease (Morita and Churg, 1983), leading to necrosis of mesangial cells and endothelial cells in glomeruli, which seriously damages glomerular function (Kengkoom and Ampawong, 2015). Glomerular diseases beginning with mesangiolysis can be recovered, of which podocytes exert a critical role in this healing process (Wnuk et al., 2012). At present, almost all western drugs can only delay the progress of CKD with great toxic and side effects, and even accelerate the development of CKD in the case of drug abuse (Olyaei et al., 2011). Meanwhile, when CKD develops to the end-stage renal failure, patients

can only maintain their life through dialysis or kidney transplantation, which is costly and extremely harmful to patients (Fabrizi et al., 2016). Traditional Chinese medicine (TCM) has been proved to have certain advantages and characteristics in the treatment of CKD (Zhong et al., 2013). Therefore, it is of great significance to seek a safe, convenient and effective TCM preparation for the treatment of CKD.

Huangkuisiwufang (HKSWF) is composed of *Abelmoschus manihot* (L.) Medik., *Astragalus mongholicus*, *Polygonum cuspidatum*, *Curcuma longa* L. *Abelmoschus manihot* (L.) Medik. has been widely used for the treatment of CKD, oral ulcers and burn in China for centuries (Committee of the Pharmacopoeia of PR China, 2010). *Abelmoschus manihot* (L.) Medik. was proved to have a vital role in anti-inflammatory, antibacterial diuresis, anti-oxidation, vasodilatation, detumescence, myocardial protection and cerebral ischemic injury (Arokiyaraj et al., 2014). *Astragalus mongholicus*, *Polygonum cuspidatum*, *Rhizoma curcuma longae* have been mainly used in folk medicine for their anti-inflammatory, immunostimulant, antioxidative, anti-cancer, antidiabetic, cardioprotective, anti-Alzheimer's disease, hepatoprotective and antiviral properties in recent years (Li et al 2014, 2016; Zhang et al., 2017). *Abelmoschus manihot* L. is widely used in the treatment of CKD, which has been developed into a new drug called Huangkui capsule. Huangkui capsule plays an important role in alleviating renal

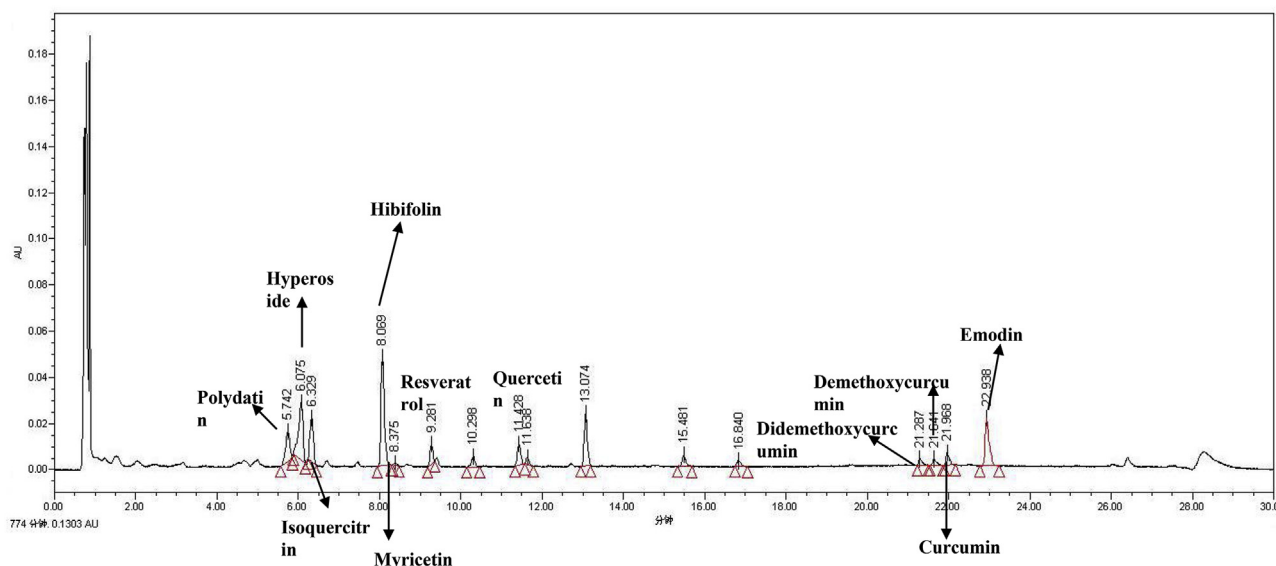


Fig. 1. Typical chemical fingerprint of HKSWF extract. The labeled compounds were identified by standard references.

inflammation, which has been illustrated to reduce renal fibrosis (Couser, 1994). Huangkui capsule is used for common CKD such as diabetic nephropathy (DN) and membranous nephropathy. Its main mechanisms include inhibiting immunity, reducing inflammation, reducing proteinuria and improving kidney function. *Astragalus* is one of the most commonly used TCMs in the treatment of CKD (Zhang et al., 2014). *Astragalus* has been proved to exhibit therapeutic effects on DN (Li et al., 2014), renal interstitial fibrosis (Zhou et al., 2017), membranous nephropathy (Leehey et al., 2010) and so on. The extract of *Polygonum cuspidatum* can reduce early renal podocyte injury (Sohn et al., 2015) and inhibit mesangial cell dysfunction (Sohn et al., 2014) in diabetic rats. *Curcuma longa* L. can attenuate renal inflammation induced by lipopolysaccharide and protect kidney function by induction of nuclear-factor-erythroid 2-related factor (Nrf2), inhibition of mitochondrial dysfunction and attenuation of inflammatory response (Zhong et al., 2009, Trujillo et al., 2013).

Therefore, we conducted this study to determine if there were some beneficial effects of HKSFW on rats with anti-Thy 1 nephritis and to explore its potential molecular mechanism.

2. Materials and methods

2.1. Huangkuisiwufang (HKSFW)

HKSFW is composed of *Abelmoschus manihot* (L.) Medik., *Astragalus mongholicus*, *Polygonum cuspidatum*, *Curcuma longa* L., which were purchased from Suzhou Tianling Chinese Herbal Medicine, Co., Ltd. and identified as qualified medicines by one of co-authors professor Jin-ao DUAN. Specimen of these four herbs have been deposited in Nanjing University of Chinese Medicine with voucher specimen number to be HKSFW20160701 (*Abelmoschus manihot* (L.) Medik.), HKSFW20160702 (*Astragalus mongholicus* (Bge.) Hsiao), HKSFW20160703 (*Polygonum cuspidatum* Sieb. et Zucc), HKSFW20160704 (*Curcuma longa* L.). The quality of the herbs and herbal extracts was consistent with the standards of Chinese Pharmacopoeia (2015). Chemical fingerprint of HKSFW was supplied as Fig. 1.

Extraction method: *Abelmoschus manihot* (L.) Medik., *Astragalus mongholicus*, *Polygonum cuspidatum*, *Curcuma longa* L. (500 g, the proportion of 10:30:15:3) were extracted two times with soaking and heating by reflux extraction method. The herbs were extracted by 7.5 L distilled water for 3 h. Then the residual herbs were extracted by 5 L 95% ethanol for 2 h. The extract solution thus obtained was combined and then subjected to drying under reduced pressure and temperature using a rotary evaporator. The proportions of these four herbs were selected by our preliminary experiment.

2.2. Phytochemical fingerprint analysis

Chromatographic analysis was performed on Acquity ultra high performance liquid chromatography (UHPLC) BEH C18 column (2.1 mm i.d. × 100 mm, 1.7 μm) using a Acquity UHPLC TM system (Waters Corp., Milford, MA, USA). The column and auto-sampler were maintained at 35 °C and 20 °C, respectively. The mobile phase, at a flow rate of 0.4 mL/min, consisted of (A) formic acid-water (1:1000, v/v) and (B) acetonitrile. The conditions of gradient eluting were optimized as the following: 10–10% B (0–6.0 min), 10–35% B (6.0–23.0 min), 35–70% B (23.0–30 min), 70–10% B (30–32.5 min), 10–10% B (32.5–35 min). The injection volume was 2 μL.

2.3. Reagents

Huangkui capsule and Enalapril (Ena) was obtained from SZYY Group Pharmaceutical Limited (Jiangsu, China). Dichloroacetate sodium (DCA) was supplied by Adamas-beta (Shanghai, China). Unless otherwise indicated, other materials and chemicals were obtained from Sigma Chemical Co. (St. Louis, MO, USA).

2.4. Animals

Male Sprague-Dawley (SD) rats (200 ± 20 g) were purchased from Beijing Vital River Laboratory Animal Technology Co., Ltd. The rats were kept in the Drug Safety Evaluation Center of Nanjing University of Chinese Medicine. All animal procedures were in accordance with government-published recommendations for the Care and Use of laboratory animals. All animal studies are in line with the guidelines of the Animal Ethics Committee of Nanjing University of Chinese Medicine (Ethics number ACU-14 (20151123)).

2.5. Model of anti-thy1-induced chronic nephritis

Model of anti-thy 1 CG was induced by injecting the monoclonal anti-Thy 1 antibody [1.75 mg/kg body wt] (Kato et al., 2013). Normal control animals were injected with equal volumes of phosphate-buffered saline (PBS) only. As a replacement of anti-Thy 1 antibody, rabbit anti-rat thymocyte serum (ATS) was produced by immunizing New Zealand Rabbit with rat's thymocyte cell suspension (Jocks et al., 1996).

2.6. Treatment with Huangkui capsule, enalapril and HKSFW

One week after modeling, rats were given Huangkui capsule in a daily dose of 750 mg/kg body wt, enalapril in a daily dose of 20 mg/kg body wt or HKSFW by gavage for 8 weeks, of which rats were given low, medium and high doses of HKSFW at 312.5, 625, 1250 mg/kg body wt. The rats in normal control (NC) group and disease control (DC) group were given the same amount of water by gavage.

2.7. Experimental groups and design

Rat urine (24 h) was collected 1 week after injecting anti-thy 1 antibody. On the basis of the 24 h proteinuria degree, forty-nine rats were randomly assigned to seven groups of 7 rats each including NC group, DC group, ambrette capsule (HC) group (positive group), enalapril (Ena) group (positive group) and HKSFW treated groups with low, middle, high doses (HZZ, HZM, HZH).

24 h rat urine was collected once weekly for 8 weeks. Rat urine was collected by metabolic cage for 24 h and urine volume was recorded. The fresh urine was centrifuged for the supernatant and stored in −20 °C refrigerator. The level of 24 h urinary protein was detected by Pierce™ BCA Protein Assay Kit (Thermo Fisher Scientific, Waltham, MA, USA). Eight weeks after treatment, the rats were anesthetized and blood was immediately collected from the aorta abdominalis of rats. BUN, Cr levels and calculated creatinine clearance (Ccr) serve as markers of renal function. Urine creatinine were measured using Quanti Chrom™ creatinine assay kit (Bio Assay System, Hayward, CA, USA) for the calculation of Ccr ratio. The detection of plasma BUN and Cr levels was determined using standard laboratory procedures. Ccr was calculated by the formula (creatinine/creatinine level) × 24Urine volume (mL)/(24 h × 60 min) (Rana et al., 2016). The kidney tissue was removed from the abdominal cavity of the rats and divided into several parts. The cortical tissue was fixed in 10% formalin for periodic acid Schiff (PAS) and was fixed in glutaraldehyde for electron microscopy for histological analysis; the remaining part was isolated, frozen in liquid nitrogen, and stored at −80 °C for protein and RNA extraction.

2.8. Cell culture

Murine mesangial cells (MMC) were donated free of charge by Nanjing Key Laboratory of Pediatrics. The cells were cultured in Dulbecco's modified Eagle's medium (DMEM; BI, Israel) of low glucose at 37 °C, 5% CO₂. All media were supplemented with 10% fetal bovine serum (FBS; BI, Israel), 100 U/mL Penicillin, 100 μg/mL streptomycin (Wang et al., 2011). The cells were grown in a 100 mm dish and were subcultured when growing to 80%–90% confluence.

The mesangial fibrosis model in cells was induced by high glucose culture, in which high glucose culture promotes the proliferation of mesangial cells, increment of inflammatory factor and collagen accumulation (Lee et al., 2018). MMC was divided into the low-glucose group (LG) cultured in the low-glucose medium (5.6 mM), the high-glucose group (HG) cultured in the high-glucose medium (25 mM), the low dose of Huangkuisiwufang group (HZL) cultured with HSKWF (50 µg/mL) in the high-glucose medium, the medium dose of Huangkuisiwufang group (HZM) cultured with HSKWF (100 µg/mL) in the high-glucose medium, the high dose of Huangkuisiwufang group (HZH) cultured with HSKWF (150 µg/mL) in the high-glucose medium, the DCA group cultured with DCA (5 mM) (Han et al., 2008) in the high-glucose medium and HSKWF + DCA group cultured with HSKWF (50 µg/mL) and DCA (5 mM) in the high-glucose medium. After intervention for 48 h, the protein was extracted from cells for western blot (WB) analysis.

2.9. Histological analysis

Electron microscopy is often used to observe the morphology of glomerular disease. The renal cortex fixed with 2.5% glutaraldehyde was made into ultrathin sections of 1 mm which is suitable for electron microscopy as previously described (Wang et al., 2015). The sections were visualized and imaged using a transmission electron microscope (Tecnai G2 Spirit Bio TWIN). Morphological changes of glomerular were studied by electron microscopy, including glomerular podocyte, structure of foot process, and glomerular basement membrane, etc.

Renal tissue was embedded in paraffin and cut into 4-µm-thick sections for morphological analysis by periodic acid Schiff (PAS) staining. Glomerular mesangial expansion was determined in 20 glomeruli per slide in a blinded manner. The images of random glomeruli were captured using an Axio VertA 1 digital camera, and the area of PAS-positive glomeruli was quantified using a computer-assisted color image analysis system (Image J). The value of positive staining area was used to evaluate the extent of glomerular matrix expansion, and scores from 0 to 4 scale (0 = 0%, 1 = 1–25%, 2 = 26–50%, 3 = 51–75%, and 4 = 76–100%) were calculated.

2.10. Western blot

Renal tissue frozen and cell precipitations were lysed in RIPA lysis buffer with protease inhibitor cocktail, and protein concentration was determined using BCA Protein Assay kit (Thermo Fisher, New York, USA). Total tissue or cell lysate mixed with sample buffer (20 µg of protein) was fractionated by SDS-PAGE on a 10% polyacrylamide gel and transferred to polyvinylidene fluoride (PVDF) membranes. After blocking with 5% skim milk at room temperature for 1 h, membranes were incubated with a primary antibody (diluted 1: 1000) overnight at 4 °C: PAI-1 antibody (ab66705, Abcam), Ang II type 1 receptor (AT1R) antibody (25343-1-AP, Proteintech), pyruvate dehydrogenase antibody (3205S, Cell Signaling Technology, Massachusetts, USA), and GAPDH (#2118, Cell Signaling Technology). Blotting of GAPDH was used as internal control. After incubating with the primary antibodies, the blots were incubated in goat anti-rabbit IgG-HRP antibody (sc-2004, Santa Cruz Biotechnology) (diluted 1: 1000) for 1 h at room temperature. The

immunostaining band was visualized using a Peiqing Science& Technology imaging densitometer.

2.11. Transcriptomic RNA-sequencing

The RNA-seq and sample analysis were provided by Xinyinzi Biotech (Shanghai, China). This project is *Rattus norvegicus* ref RNAseq data sequencing and analysis technology service. After the total RNA was extracted from rat kidney, the mRNA was enriched with magnetic beads with Oligo (dT). After constructing RNA sequencing library, HiSeq X-Ten was utilized for PE150 sequencing. After strict data quality control, data were analyzed and differentially expressed genes were found.

2.12. Statistical analysis

Results are expressed as mean ± standard deviation (SD). Student's *t*-test was used to determine significance differences between two groups and one-way ANOVA was used to compare more than two groups. *P* value < 0.05 was considered significant.

3. Results

3.1. Phytochemical fingerprint of the HSKWF

Eleven components (polydatin, hyperoside, isoquercitrin, hibifolin, myricetin, resveratrol, quercetin, didemethoxycurcumin, demethoxycurcumin, curcumin and emodin) were detected in HSKWF which were confirmed by standard references. Compound name of each peaks were also inserted in Fig. 1.

3.2. HSKWF improves renal function in rat Thy-1 nephritis

We initially observed that the levels of plasma BUN and Cr were significantly increased (*P* < 0.05) and the Ccr levels were significantly decreased (*P* < 0.05) in the disease rats compared with normal rats. Meanwhile, we found that the levels of plasma BUN were dramatically decreased (*P* < 0.05) after treatment with HSKWF, of which the decrease of approximately 27.1% was detected in HZL group compared with DC group. Although the Ccr levels after HSKWF treatment appeared to be rising, there was no statistical significance (Table 1). As shown in Fig. 2, total proteinuria was obviously elevated (*P* < 0.05) after ATS injection and increased during the following 8 weeks. Treatment with HSKWF showed a remarkable reduction of proteinuria (*P* < 0.05) in a dose-dependent manner, and the level of proteinuria was lower in rats with low dose of HSKWF than the other two groups (HZM and HZH) (Fig. 2). These results imply that HSKWF inhibits urinary protein excretion and improves renal function of anti-Thy 1 nephritis rats.

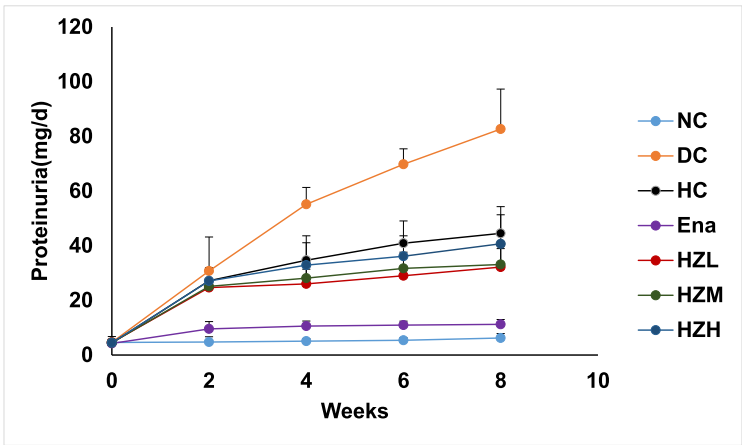
3.3. HSKWF alleviates glomerular injury in rat Thy-1 nephritis

Compared with NC group, DC group showed typical characteristics of glomerulus damage: mesangial cells proliferation, mesangial matrix hyperplasia, basement membrane thickening slightly and fusion of foot

Table 1
Renal function of the experimental rats.

Renal Function	NC	DC	HC	Ena	HZL	HZM	HZH
Plasma BUN(mmol/L)	5.31 ± 0.16	8.53 ± 1.67*	6.97 ± 0.41	6.01 ± 0.74 [#]	6.22 ± 0.49 [#]	6.30 ± 0.48 [#]	6.32 ± 0.35 [#]
Plasma creatinine (umol/L)	29.5 ± 0.93	37.33 ± 2.07*	31.67 ± 2.50 [#]	35.83 ± 4.31	35.50 ± 2.07	33.50 ± 3.08	35.67 ± 7.92
Creatinine clearance (ml/min)	2.38 ± 0.93	1.32 ± 0.57*	1.60 ± 0.77	2.04 ± 0.94	1.70 ± 0.22	1.55 ± 0.27	1.56 ± 0.70

NC: normal control; DC: disease control; HC:ambrette capsule(750 mg/kg/d); Ena: enalapril(20 mg/kg/d); HZL: Huangkuisiwufang (312.5 mg/kg/d); HZM: Huangkuisiwufang (625 mg/kg/d); HZH: Huangkuisiwufang (1250 mg/kg/d). **p* < 0.05 vs NC; [#]*p* < 0.05 vs DC.



P value	NC	DC	HC	Ena	HZL	HZM	HZH
vs NC		0.015	0.010	0.015	0.006	0.006	0.007
vs DC			0.277	0.023	0.126	0.145	0.221

Fig. 2. HKSWF decreased 24-h urinary protein excretion in rats with anti-Thy 1 nephritis.

processes. HKSWF-treated groups improved basal membrane thickening, mesangial cell and epithelial cell proliferation and foot process fusion to varying degrees (Fig. 3A). Fig. 3B and C illustrated the thickness of the glomerular basement membrane and the width of the foot processes, respectively, and HKSWF-treated rats showed an evident reduction of glomerular basement membrane thickness compared with

DC rats, of which the decrease was approximately 30.8% in HZM group. Meanwhile, the width of the foot processes was significantly decreased after treated with HKSWF, of which low dose exhibited the best effect with the decrease of approximately 18.4% ($P < 0.05$) compared with DC group (Fig. 3B & C). These results indicate that HKSWF improves glomerular injury in rats with anti-Thy 1 nephritis.

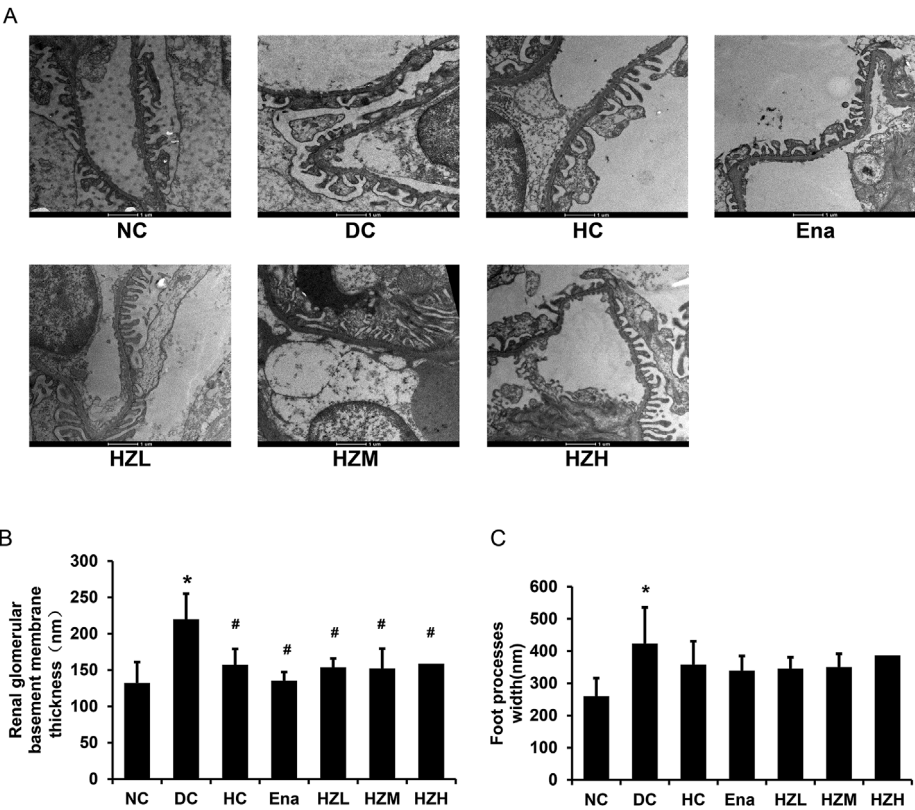


Fig. 3. Effects of HKSWF on improving podocyte injury in experimental animals. (A) The pathological changes of renal cortex were observed by electron microscopy (6800 $\times$). (B) The scores of renal glomerular basement membrane thickness. (C) The scores of foot processes width. * $P < 0.05$ vs NC; # $P < 0.05$ vs DC.

3.4. HKSWF attenuated oxidative stress in rats with anti-Thy 1 nephritis

It has been reported that Angiotensin II (Ang II) increases NAD(P)H oxidase and promotes reactive oxygen species (ROS) production via AT1R (Rincon et al., 2015). Thus, AT1R can be regarded as an indicator to evaluate the level of oxidative stress in the kidney. Here we evaluated the expression of AT1R after modeling and drugs administration by western blot. As a result, anti-Thy1 nephritis model presented higher protein expression of AT1R than normal controls ($P < 0.05$) (Fig. 4). The HKSWF treatment at high and middle doses have a remarkable inhibitory effect on the expression of AT1R ($P < 0.05$). These results showed that HKSWF may abolish renal oxidative stress by inhibiting AT1R protein expression (Fig. 4).

3.5. HKSWF ameliorates renal fibrosis in rat Thy-1 nephritis

Taking advantage of PAS staining, we found that the proliferation of mesangial cells, mesangial matrix, and glomerular cyst wall were increased in DC group. Meanwhile, some renal tubular epithelial cells were swollen, with the lumen narrowed, and inflammatory cells infiltrated into the stroma. Indeed, HKSWF treatment attenuated the histopathological changes of renal tissue in anti-Thy 1 nephritis rats (Fig. 5A). Glomerular matrix expansion was evaluated by the glomerular scores in PAS staining. As shown in Fig. 5B, glomerular matrix expansion significantly increased in DC rats compared with NC group. Treatment of HKSWF produced remarkable decrease in glomerular matrix expansion, and the most significant decrease of approximately 41.3% was observed in HZL group ($P < 0.05$) (Fig. 5B).

Plasminogen activator inhibitor-1 (PAI-1), proved to be involved in renal fibrosis, protects extracellular matrix (ECM) proteins from degradation (Ghosh and Vaughan, 2012). PAI-1 was detected by WB analysis, and we found that the expression of PAI-1 in the kidney of DC group was significantly higher by 29.0% than that of the NC group ($P < 0.05$). Meanwhile, the expression of PAI-1 was inhibited by HKSWF treatment (Fig. 5C).

Taken together, these above data indicated that the treatment with HKSWF has positive effects on kidney function, renal matrix accumulation and podocyte injury in anti-Thy1-induced chronic glomerular nephritis.

3.6. RNA-seq reveals the potential signaling pathway of HKSWF in the treatment of CG

To further investigate the underlying mechanism of the HKSWF nephroprotective effect, we performed transcriptomic mRNA expression profiling. The differentially expressed genes (DEGs) among NC group, DC group, HC group and HKSWF treated group were obtained and analyzed. We screened out 322 significantly up-regulated genes and 270 significantly down-regulated genes between HZM and DC groups (Fig. 6A). The Venn diagram showed that 188 DEGs were significantly altered in all four groups (HC, HZL, HZM, HZH) compared with DC group (Fig. 6B). Furthermore, we identified that the three most significantly enriched Gene Ontology (GO) terms for biological process (BP), cellular component (CC) and molecular function (MF) were related to metabolic process, and organelle and catalytic activity, respectively (Fig. 6C). Only the top 20 most significantly enriched pathways were shown in Fig. 6D. The KEGG analysis clarified the most significantly upregulated pathway was Pyruvate metabolism in HKSWF group compared to DC group (Fig. 6D). Based on the identified mRNA regulation, the Pyruvate metabolism signaling pathway was found to be one of the most significantly enriched pathways. These findings imply that HKSWF may protect kidney by regulating Pyruvate metabolism pathway.

3.7. Pyruvate metabolism signaling pathway is involved in the treatment of HKSWF in vivo and in vitro

Consistent with the results of RNA-seq, pyruvate dehydrogenase was detected by western blot analysis in kidney tissue of rats with glomerulonephritis, and the expression of pyruvate dehydrogenase was evidently decreased by 60.9% ($P < 0.05$) in rats treated with low dose of HKSWF compared to DC group (Fig. 7A). To further investigate the role of pyruvate metabolism signaling pathway in the treatment of chronic nephritis by HKSWF, the expression of pyruvate dehydrogenase in MMCs were detected, and we found that HG-induced up-regulation of pyruvate dehydrogenase was reversed by HKSWF (Fig. 7B), of which high dose exhibited the best effect with the decrease of approximately 42.6% compared with HG group ($P < 0.05$). In addition, PAI-1 related with renal fibrosis was detected and we found HKSWF reversed the up-regulation of PAI-1 expression induced by HG culture, of which the significant decrease was 11.6% observed in HZL group (Fig. 7C). Furthermore, as an agonist of pyruvate dehydrogenase, DCA increased the expression of PAI-1, as well as reversed the inhibitory effect of HKSWF on PAI-1 by 14.5% (Fig. 7D).

These results demonstrated that HKSWF can decrease the expression of pyruvate dehydrogenase and activate pyruvate metabolism pathway.

4. Discussion

As the leading cause of end-stage kidney disease, CG is a slow progressive decline in renal function, which could be a serious risk of people's health (Floege and Amann, 2016). The clinical manifestations of this disease are hematuria, proteinuria, edema and hypertension (Guo et al., 2010). The progression of glomerulonephritis is accompanied by renal fibrosis, which gradually leads to a progressive process of renal function from health to injury, to damage, to loss of function (Nogueira et al., 2017). Renal fibrosis is characterized by abnormal deposition of ECM (Harendza et al., 1999). Nephritis belongs to edema, asthenia, low back pain, dizziness and other categories of TCM, and the etiology and pathogenesis of chronic nephritis are quite complicated (Shi et al., 2012). Modern medical treatment programs have great side effects and are prone to relapse. TCM or combination of traditional Chinese and western medicine treatment is expected to improve the efficacy of chronic nephritis. HKSWF is a complex of four traditional

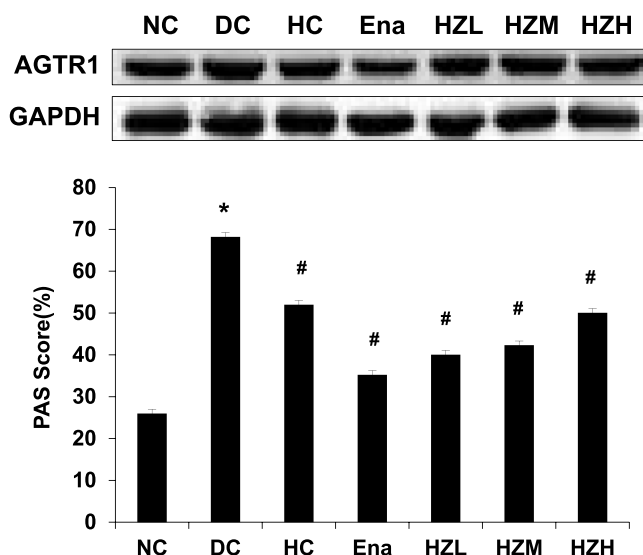


Fig. 4. Effects of HKSWF on improving oxidative stress in experimental animals. The level of AT1R protein was shown by Western blot. The relative expression level of AT1R protein was measured by using GAPDH densitometric intensity as control. * $P < 0.05$ vs NC; # $P < 0.05$ vs DC.

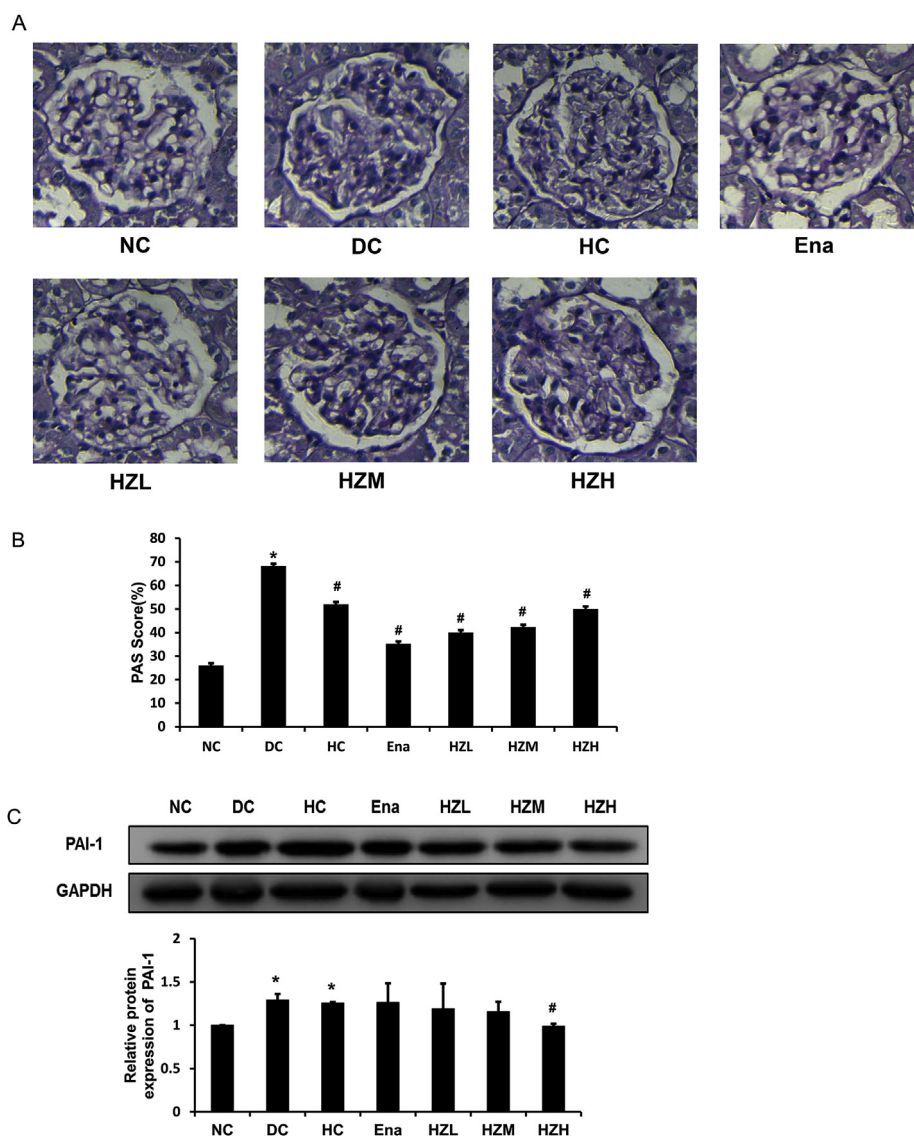


Fig. 5. Effects of HKSWF on improving renal fibrosis in experimental animals. (A) Glomeruli of kidney stained with Periodic Acid-Schiff stain (PAS) (400 x). (B) The score of glomerular extracellular matrix. (C) The level of PAI-1 protein expression analyzed by Western blot. The relative expression level of PAI-1 protein was measured by using GAPDH densitometric intensity as control. * $P < 0.05$ vs NC; # $P < 0.05$ vs DC.

Chinese medicines, *Abelmoschus manihot* (L.) Medik., *Astragalus mongholicus*, *Polygonum cuspidatum*, *Curcuma longa* L. The ambrette seed oil has been shown to be a source of antimicrobial agents due to its antimicrobial activity (Arokiyaraj et al., 2014). A previous study showed that *Astragalus mongholicus* treated IgA nephropathy by upregulating Cosmc expression in B lymphocytes and reversing the levels of aberrant O-glycosylation (Ji et al., 2014). Furthermore, a clinical study highlights that the *Astragalus mongholicus*-based Chinese Medicine Formula has the potential in preserving residual renal function of patients undergoing peritoneal dialysis (Lui et al., 2015). The role of *Astragalus mongholicus* in the treatment of membranous nephropathy has been documented to be related to the attenuation of podocyte injury (Chen et al., 2014). *Polygonum cuspidatum* extract has been administered to prevent streptozotocin-induced diabetic rats from proteinuria and podocyte loss (Sohn et al., 2015). Curcumin, a compound extracted from *Curcuma longa* L., has been shown to be a potential renoprotective molecule in several experimental models of nephritis (Trujillo et al., 2013); moreover, Curcumin attenuates LPS-induced experimental renal inflammation by inhibiting the expression of MCP-1 mRNA and DNA-binding activity of NF- κ B (Fang et al., 2011).

Anti-Thy1 nephritis model is an animal model of experimental

mesangial proliferative glomerulonephritis, which is induced by the injection of rabbit anti-rat thymocyte serum in SD rats. In the pathogenesis, the immune response activates inflammatory cells and makes them release inflammatory mediators, which results in inflammatory reactions, glomerular damage (damage of membrane molecular barrier and charge barrier), and the appearance of proteinuria (Ikezumi et al., 2002). As indicators to evaluate renal function, BUN and Cr were increased and Ccr was significantly decreased in disease group compared with normal group, which is consistent with the previous reports (Zhang et al., 2018). Meanwhile, anti-Thy 1 nephritis model rats had a significant increase in 24 h urinary protein excretion, and their pathological sections showed severe kidney injury. It has been documented that enhanced oxidative stress is involved in initiation and progression of Thy-1 nephritis model (Nakamichi et al., 2007). Our study confirmed that AT1R expression was remarkably upregulated after modeling. In addition, PAI-1 expression associated with renal fibrosis was obviously elevated in disease group. Proteinuria is a marker of glomerular injury and an independent risk factor for the progression of CKD, therefore, lowering proteinuria may delay disease progression (Goriz and Martinez-Castelao, 2012). The results showed that HKSWF was significantly superior to Huangkui capsule in reducing 24 h urinary

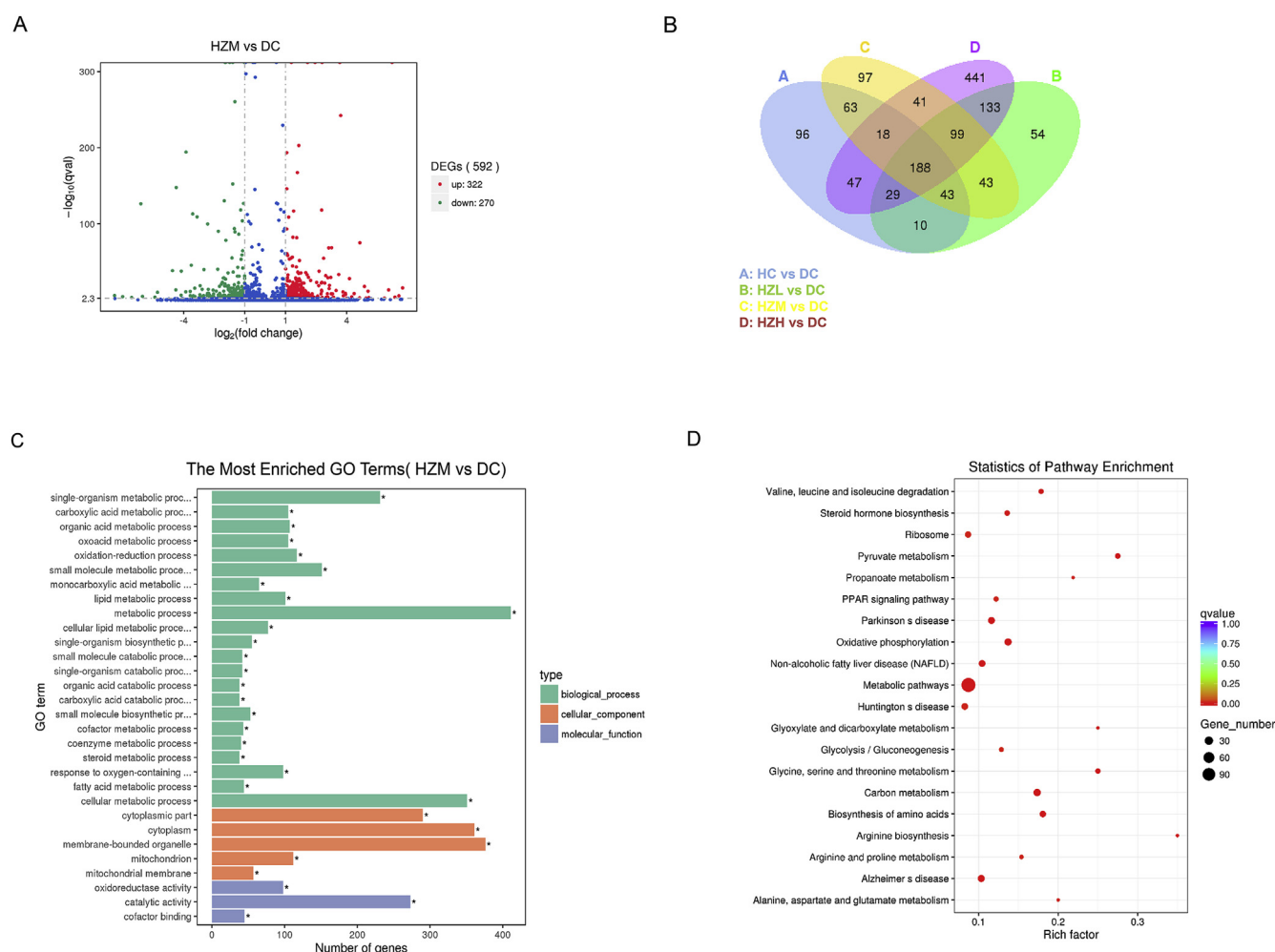


Fig. 6. Large scale gene expression profiles between NC, DC and HKSWF groups. (A) The volcano plot is used for enabling visualization of the relationship between fold change and statistical significance. The red point in the plot represents upregulated RNA (fold change > 2 , $P < 0.05$) and the green point represents down-regulated RNA (fold change < 2 , $P < 0.05$), while the blue point represents RNA with no statistical differences (fold change < 2 , $P > 0.05$). (B) The Venn diagram of differential expressed genes detected in the four different groups. The overlap represents 20 differentially expressed gene common to the three different groups. (C) GO (Gene ontology) classification of target genes. The 30 most significantly enriched GO terms for whole kidney. The x-axis shows the numbers of gene and y-axis shows the diverse biological functions of target genes according to the three GO categories including biological process, cellular component, molecular function. (D) The 20 most significant up-regulated pathways (HZM VS DC). KEGG (Kyoto Encyclopedia of Genes and Genomes) pathway classification of target genes. Scatter diagram of enriched KEGG pathways of 20 differentially expressed gene. The Y-axis represents the name of pathway and the X-axis represents the Rich factor. The point size indicates the number of differentially expressed genes in one pathway, and the color of the point corresponds to the range of the Q value. NC: normal control group; DC: disease control group; HZL, HZM, HZH: HKSWF treated groups with low, middle, high doses. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

protein excretion and BUN, improving oxidative stress and ECM dilation, alleviating renal injury and renal fibrosis. Podocytes, attached to the lateral surface of glomerular basement membranes, are the important components of the glomerular blood filtration barrier, and their damage can seriously affect the glomerular function (Pavenstädt, 2000). Interestingly, it was also found that the low-dose and medium-dose HKSWF had a higher protective effect on rats' kidneys than the high-dose, suggesting that there must be an optimal dosage for the protective effect. The dose of 312.5 mg/kg/d had the best therapeutic effect on nephritis in this study. However, the optimal dose of HKSWF for nephritis treatment needs to be further explored.

According to the results of RNA sequencing, we speculated that pyruvate metabolism was the potential signaling pathway in the treatment of CG with HKSWF. To further investigate the mechanism underlying the role of HKSWF in nephritis treatment, the expression of pyruvate dehydrogenase and PAI-1 in MMCs were detected by western blot analysis, and the results indicates that HKSWF inhibits the expression of these two proteins. DCA was used extensively to promote

pyruvate dehydrogenase complex (PDC) activity in the laboratory and clinically (Michelakis, 2008). In this study, DCA reversed the decrease of PAI-1 protein induced by HKSWF, suggesting that HKSWF may treat nephritis by declining pyruvate dehydrogenase signaling.

Pyruvate, a tricarboxylic acid produced in the body, is the final product of glycolysis pathway. It is reduced to lactic acid for energy supply in the cytoplasm, or oxidized into acetyl CoA in the mitochondria to enter the cycle of tricarboxylic acid, which is oxidized into carbon dioxide and water to complete the aerobic oxidation of glucose for energy supply (Gray et al., 2014). Therefore, pyruvate plays an important role in the energy metabolism of the body. The metabolic imbalance of pyruvate will seriously affect the normal function of body, leading to the occurrence of diseases such as cancer (Rauckhorst and Taylor, 2016) and neurodegeneration (Ullah et al., 2013). Recent studies found that both ischemic renal injury and nephrotoxic renal injury lead to significant and sustained depletion of pyruvate, and pyruvate therapy may reduce renal injury, which may be due to the decrease of inflammation and preserve of the enzymatic machinery essential for

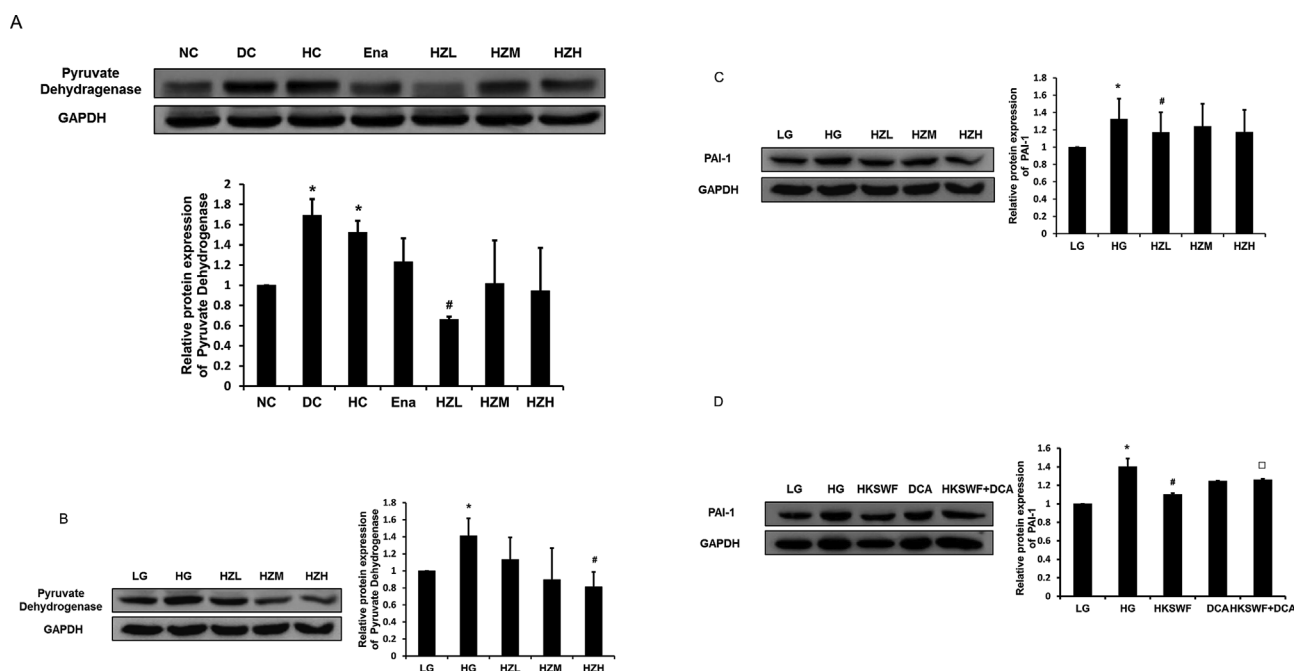


Fig. 7. Western blotting indicating that pyruvate metabolism was inhibited by HKSWF *in vivo* and *in vitro*. (A) Expression of pyruvate dehydrogenase in renal cortex. (B–D) The expressions of pyruvate dehydrogenase and PAI-1 in MMCs. The relative expression level of PAI-1 or Pyruvate Dehydrogenase protein were measured by using GAPDH densitometric intensity as control. LG: low glucose, HG: high glucose, HZL/HKSWF: Huangkuisiwufang (50 μ g/mL), HZM: Huangkuisiwufang (100 μ g/mL), HZH: Huangkuisiwufang (150 μ g/mL), DCA: Dichloroacetate sodium (5 mM). * $P < 0.05$ vs NC/LG; # $P < 0.05$ vs DC/HG; □ $P < 0.05$ vs HKSWF.

cell viability (Gregorini et al., 2017; Johnson and Zager, 2014). Pyruvate dehydrogenase kinase 4 may be a therapeutic target for acute kidney injury for its inhibitory effect on cisplatin-induced acute renal injury (Oh et al., 2017). These findings suggest that pyruvate metabolism is strongly associated with kidney injury. In this study, we found HKSWF decreases the expression of pyruvate dehydrogenase and improves nephritis. Future studies may focus on the molecular mechanisms underlying the inhibitory effect of HKSWF on pyruvate metabolism pathway.

5. Conclusions

Our study demonstrates that HKSWF presents therapeutic effect on kidney in anti-Thy 1 nephritis model. The potential mechanism of HKSWF function is mediated through attenuating pyruvate dehydrogenase to improve glomerular injury. These findings suggest that HKSWF could be a promising therapeutic agent for the treatment of glomerulonephritis.

Author's contributions

Chunyan Gu, Jin-ao Duan and Jianming Guo designed and supervised this study. Ting Lu and Chunyan Gu collected and analyzed data, wrote and edited the manuscript. Ting Lu, Yong Bian and Mengjie Guo performed the experiments, collected and analyzed the data, generated the figures, and wrote the manuscript. Ye Yang and Jianming Guo provided technical counseling on experiments and edited the manuscript. Chunyan Gu and Jin-ao Duan reviewed the data and edited the manuscript.

Disclosure of potential conflicts of interest

No financial or commercial conflict of interest exists.

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